Exploratory Analysis of Electricity Consumption Behaviour in Student Hostels

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Abstract— Electricity consumption in student hostels is influenced by occupancy, academic schedules, weather conditions, holidays, and the use of electrical appliances. Aggregate meter readings alone often provide limited insight into the reasons behind demand variation, peak usage, and unusual consumption. This paper presents an exploratory data analysis (EDA) framework for examining electricity-consumption behaviour in student hostels. The framework includes data validation, missing-value analysis, descriptive statistics, temporal pattern identification, contextual comparisons, and statistical outlier detection. It considers occupancy, academic schedules, weather, holidays, and hostel-block variations only when the corresponding variables are available in the dataset. The study also discusses the limitations of aggregate measurements and the need to combine energy records with contextual institutional data. The proposed analysis is intended to support evidence-based understanding of hostel energy demand and provide a foundation for future energy-management studies.

Index Terms— Student hostels, electricity consumption, exploratory data analysis, occupant behaviour, energy management, peak demand, outlier detection, building energy.

I. INTRODUCTION

Electricity is a necessary resource in student hostels, where lighting, fans, air conditioners, water heaters, computers, charging devices, and shared facilities operate throughout the day. Unlike many residential settings, hostels have changing occupancy patterns shaped by class hours, examinations, weekends, semester breaks, and holidays. These factors may cause electricity demand to rise or fall at different times.

Understanding these patterns is important for campus facility management. High consumption during particular hours may indicate simultaneous use of cooling equipment, lighting, or other appliances. A sudden increase may also arise from changes in occupancy, equipment operation, or measurement errors. Without systematic analysis, it is difficult to distinguish normal variations from readings that deserve investigation.

Traditional aggregate meter readings can show the total electricity consumed by a building, but they may not explain the causes of consumption. A time-based analysis can identify when demand is high, while contextual information can help examine whether occupancy, temperature, or academic schedules are associated with the observed variation. Block-level comparisons may also help identify buildings with different demand profiles, provided that comparable meter information is available.

This paper therefore focuses on exploratory data analysis rather than claiming a completed prediction system. It proposes a structured process for inspecting hostel energy records, identifying temporal patterns, examining available contextual variables, and detecting unusual observations. The study treats correlation as descriptive evidence and does not interpret it as proof of causation.

The main contributions are: (1) a practical EDA workflow for hostel electricity records; (2) analysis of hourly, daily, and weekday consumption patterns; (3) a conditional approach to examining occupancy, weather, holidays, academic schedules, and block differences; and (4) a discussion of data requirements and limitations for reliable hostel energy analysis.

II. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement: Electricity usage in student hostels varies according to occupancy, academic schedules, weather, holidays, and individual usage habits. Institutions often have only aggregate consumption information, making it difficult to understand unusual usage patterns, periods of excessive consumption, and variations between hostel blocks.

The overall objective is to explore available electricity-consumption records and identify patterns that can be supported by the data. The specific objectives are:

1) To inspect the dataset for missing values, duplicate records, inconsistent data types, and unusual observations.

2) To summarise electricity consumption using descriptive statistics and visualisations.

3) To identify hourly, daily, weekday, and weekend usage patterns.

4) To examine occupancy and weather relationships when suitable time-aligned variables are available.

5) To compare holiday, examination, and regular academic periods when calendar labels are provided.

6) To compare hostel blocks using consistent meter coverage and building identifiers.

7) To detect potential high-usage and low-usage outliers for further inspection.

III. RELATED WORK

Building energy research has increasingly recognised that occupant behaviour contributes to variations in electricity demand. Chen et al. [2] reviewed the impact of occupant behaviour on building energy consumption and discussed the importance of understanding interactions between occupants and building systems. Uddin et al. [3] examined modelling and simulation approaches for occupant behaviour and energy conservation, showing the relevance of behavioural information in energy studies.

Paone and Bacher [4] reviewed methods for understanding and influencing occupant behaviour to improve building energy efficiency. Their work highlights that energy-use patterns are not determined only by building characteristics; the way occupants use facilities also matters. Bäcklund et al. [5] focused on campus buildings and reviewed how occupant behaviour can influence energy use in educational environments.

The dormitory dataset described by Pandey et al. [1] is particularly relevant to the present topic because it combines occupant behaviour, indoor environmental information, and energy-use observations around a dormitory retrofit. Krishnan et al. [6] reviewed data-driven occupant behaviour models for assessing demand-side energy consumption. Xu et al. [7] discussed research directions in occupant energy-consumption behaviour, including the need for richer data and wider building coverage.

These studies motivate an analysis that considers temporal energy patterns together with occupancy and environmental context. However, the presence of a variable in related research does not mean it is available in the dataset used here. All conclusions in the present study must be restricted to verified dataset fields.

IV. DATASET DESCRIPTION

The intended reference dataset is “Dataset on occupant behavior, indoor environment, and energy use before and after dormitory retrofit” [1]. It provides a dormitory-related context for investigating energy use and occupant/environmental information. Before analysis, the downloaded CSV must be inspected to confirm the exact file, number of records, sampling interval, units, variable names, and data coverage.

The core data required for temporal EDA are a timestamp and one or more electricity or energy measurements. Additional variables may include occupancy, indoor temperature, humidity, illumination, carbon dioxide, outdoor weather, holiday indicators, academic schedule labels, and hostel or building identifiers. These optional fields must be confirmed from the actual file.

If multiple energy channels are present, their measurement boundaries must be checked before combining them. For example, a whole-building meter and a sub-circuit meter may overlap. Adding overlapping readings would double-count consumption and produce an incorrect total. Similarly, power readings and energy readings are different quantities and should not be combined without an appropriate time-based conversion.

Dataset coverage and suitability

The selected dormitory dataset is relevant because it connects occupant-related observations with indoor environmental conditions and energy use. Before analysis, the downloaded files should be inspected to identify the exact meter, sensor, and occupant variables supplied. The study should retain the original measurement units and sampling intervals, and should document any aggregation applied to the readings. Where several files represent different rooms, floors, or monitoring periods, their identifiers should be preserved so that records can be compared without losing their source context.

The availability of a variable must be confirmed from the data dictionary and the CSV columns. If the dataset does not provide hostel-block labels, resident counts, academic-calendar events, or outdoor weather, those factors cannot be tested directly in this paper. They should be reported as unavailable rather than estimated or assumed. Indoor temperature and humidity, when present, describe the monitored indoor environment and should not automatically be treated as outdoor weather measurements.

For this reason, the analysis separates directly observed variables from contextual factors proposed for future data collection. The primary EDA will use only verified energy and environmental measurements. Additional occupancy and schedule comparisons can be performed if matching records are obtained from hostel administration. This approach keeps the conclusions traceable to the available evidence and avoids claiming relationships that the dataset cannot establish.

TABLE I. DATA FIELDS REQUIRED FOR THE PROPOSED ANALYSIS

Field group	Example information	Purpose
Time	Timestamp / date / hour	Temporal usage patterns
Energy	kWh or power readings	Consumption and peak demand
Occupancy	Residents / people count	Occupancy–usage relationship
Weather	Temperature / humidity	Environmental relationship
Calendar	Holiday / exam / class day	Schedule comparisons
Building	Hostel / block / wing ID	Block-wise comparison

The dataset may not contain all fields listed in Table I. Academic-calendar labels, official holiday flags, occupancy counts, and hostel-block identifiers must not be assumed. If absent, they should be collected from institutional records and aligned with the energy timestamps. This distinction is important because aggregate energy records alone cannot establish the reasons behind every consumption change.

V. PROPOSED METHODOLOGY

A. Data Loading and Initial Inspection

The CSV file is loaded using Python and Pandas. The first stage records the number of rows and columns, column names, data types, sample observations, and descriptive statistics. Timestamp fields are converted into a consistent date-time format. Energy columns are checked for their units and converted to numeric values where appropriate. Identifier columns that do not contribute to descriptive analysis are retained for grouping but should not be treated as numerical measurements.

B. Data Cleaning and Missing-Value Analysis

Missing values are counted for each variable and expressed as a percentage of the total records. A missing-value heatmap can show whether gaps are isolated or occur in continuous periods. Duplicate rows are identified and removed only after confirming that they are repeated observations. Invalid timestamps, negative readings, impossible values, and inconsistent units are examined before further analysis.

The treatment of missing values depends on the measurement type. Short gaps in a time series may require a different approach from long periods of missing readings. Missing occupancy or calendar information should not be replaced with arbitrary values because this may create false relationships. Each cleaning decision should be documented.

C. Descriptive and Distribution Analysis

Mean, median, minimum, maximum, standard deviation, and quartiles are calculated for available energy variables. Histograms help identify the shape of the consumption distribution, while boxplots show spread and potential extreme values. The difference between mean and median can indicate skewness caused by occasional high-consumption observations.

D. Temporal Consumption Analysis

When a valid timestamp is available, energy readings are grouped by hour, date, and day of the week. An hourly profile can reveal periods of comparatively high or low average usage. Daily aggregation can show changes over the observation period, while weekday comparisons can reveal recurring differences between working days and weekends. The measurement interval must be considered when calculating daily totals.

E. Occupancy and Academic-Schedule Analysis

If student counts or occupancy measurements are available, they can be aligned with energy readings using matching timestamps. Scatter plots and correlation measures can be used to explore whether higher occupancy is associated with higher consumption. Academic schedule labels may be used to compare class days, examination periods, weekends, and semester breaks. These comparisons require accurate calendar information; date alone does not reliably identify an examination or holiday.

F. Weather and Environmental Analysis

Temperature and humidity may be examined against electricity use using scatter plots, grouped averages, and correlation analysis. For example, higher outdoor temperatures may coincide with increased cooling demand, but this relationship depends on building equipment and operating practices. Indoor environmental variables such as temperature, humidity, or illumination can provide additional context where measured. Observed associations should not be described as causal effects without further investigation.

G. Hostel-Block Comparison

If the dataset contains a hostel or building identifier, consumption can be grouped by block. Mean, median, maximum, and total energy use can be compared only when meter coverage, units, sampling intervals, and observation periods are consistent. Where building sizes or occupancy differ, per-capita or floor-area-normalised comparisons may be more informative, provided the required denominator data are available.

H. Outlier Detection

The Interquartile Range method is used to flag potential outliers. For first quartile Q1 and third quartile Q3, IQR = Q3 − Q1. Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are flagged for review. These records may represent real peak demand, unusual operating conditions, sensor errors, or data-entry problems. Outliers should be investigated rather than automatically deleted.

I. Visualisation and Interpretation

The proposed visual outputs include missing-value heatmaps, histograms, boxplots, hourly profiles, daily trend plots, weekday comparisons, occupancy and weather scatter plots, block-wise charts, and correlation heatmaps. Each figure should have a clear title, labelled axes, and units. Interpretation should describe the visible pattern, state the data coverage, and avoid conclusions that are unsupported by the measurements.

VI. RESULTS AND DISCUSSION

This paper describes the EDA framework and the analyses to be carried out. Verified numerical findings are not included because the final dataset outputs have not been provided for validation. Values such as average consumption, peak hour, maximum demand, outlier count, correlation coefficients, and block rankings must be taken directly from the executed notebook rather than estimated.

After execution, the results should first describe the final dataset dimensions, measurement interval, energy units, missing-value distribution, and number of duplicate records removed. The temporal analysis should then identify the periods with the highest and lowest measured usage. If occupancy, weather, calendar, or block information is present, the corresponding comparisons should be reported with the number of valid observations used.

The discussion should distinguish between measured facts and possible explanations. A high-consumption period may coincide with high occupancy or hot weather, but this does not establish that either factor caused the increase. Likewise, an IQR outlier is a statistical flag and not proof of electricity wastage. Facility-level inspection or additional sensor information may be required to interpret unusual readings.

VII. LIMITATIONS

The main limitation is dependence on the exact variables and measurement coverage of the selected dataset. A dormitory dataset collected at another institution may not represent the climate, building design, occupancy patterns, or energy practices of the authors' campus. Aggregate readings cannot independently identify individual appliance use or personal student habits.

If occupancy counts, academic-calendar labels, holiday records, weather observations, or hostel-block identifiers are absent, those parts of the problem statement cannot be directly evaluated. A second limitation is that meter readings may cover different circuits or time intervals, making direct comparisons misleading. Potential outliers also require contextual review because unusually high values may be legitimate peaks rather than errors.

VIII. CONCLUSION AND FUTURE WORK

This paper presents a structured exploratory framework for understanding electricity-consumption behaviour in student hostels. The proposed process includes data validation, missing-value analysis, descriptive statistics, temporal pattern identification, contextual comparisons, and outlier detection. It provides a systematic way to inspect when electricity use is high, how consumption varies over time, and which records may require further investigation.

The framework also emphasises that reliable explanations require more than aggregate meter readings. Occupancy counts, academic schedules, holiday calendars, weather observations, and hostel-block identifiers should be aligned with energy measurements wherever possible. Without these fields, the analysis must remain limited to the temporal and statistical patterns that the dataset directly supports.

Future work will execute the EDA on the validated dataset and incorporate the actual charts and quantitative findings. Additional institutional data may be integrated to examine occupancy-adjusted demand, examination and holiday effects, and differences between hostel blocks. More detailed circuit- or appliance-level monitoring may help identify the sources of peak demand and support practical energy conservation measures.

IX. ADDITIONAL ANALYSIS AND INTERPRETATION

A useful EDA study should move beyond plotting raw meter values and examine whether the observed patterns are stable, comparable, and meaningful. The first step is to establish the measurement interval. Hourly, 15-minute, and daily readings represent different levels of detail; comparisons should therefore be made only after the timestamps have been interpreted correctly. If the source contains cumulative meter readings, interval consumption should be derived from consecutive readings after checking for meter resets and negative differences. If the source already contains interval energy, it should not be differenced again.

A. Normalising consumption

Where a reliable occupancy count is available, energy use may be expressed per resident or per occupied bed for the corresponding time interval. This helps distinguish an increase caused by more residents from an increase in usage intensity. The denominator must represent the same time period as the energy measurement. Empty rooms, missing occupancy records, and changes in resident count should be treated explicitly; dividing by an assumed hostel capacity can produce misleading comparisons when actual occupancy is unknown.

B. Comparing time periods

Hourly profiles can reveal recurring morning, afternoon, and night-time demand patterns. Daily summaries can be compared across weekdays and weekends, while monthly summaries can help identify longer-term changes. These comparisons should account for incomplete days and unequal observation periods. A day with only a few recorded intervals must not be treated as a complete low-consumption day. Where academic-calendar labels are available, examination periods, regular teaching days, and vacation periods can be compared as separate groups rather than being inferred from dates alone.

C. Interpreting unusual observations

An outlier is a record that differs substantially from the distribution of the data; it is not automatically evidence of wastage or equipment failure. A sudden increase may reflect a genuine high-demand event, a change in occupancy, a special activity, a meter-reading issue, or a change in measurement configuration. Each flagged record should therefore be checked against neighbouring timestamps, missing intervals, meter boundaries, and available contextual information. The original observation should be retained, and any exclusion or correction should be documented with a reason.

D. Linking patterns with possible causes

Relationships between energy use and temperature, humidity, occupancy, or illumination can be explored using scatter plots and correlation measures when these variables are present. Correlation indicates association, not causation. For example, higher outdoor temperature may coincide with increased cooling demand, but the conclusion requires information about cooling equipment, indoor conditions, and operating schedules. Similarly, a weekend reduction cannot be attributed to lower occupancy unless occupancy data support that interpretation.

X. PRACTICAL IMPLICATIONS FOR HOSTEL MANAGEMENT

A verified consumption profile can help facility teams identify periods that deserve closer inspection. Repeated night-time base loads may justify checking common-area lighting, water pumps, and equipment schedules. Short-duration peaks may require inspection of high-power appliances or simultaneous equipment operation. These are investigation priorities, not conclusions that can be drawn from an aggregate meter alone. Metering at block or circuit level would make the source of demand easier to locate.

The analysis can also support a simple monitoring routine. Management may review daily totals, compare them with a recent baseline, and record operational events that could explain large deviations. Any alert threshold should be selected using the historical variability of the particular building and reviewed as occupancy or operating conditions change. A single fixed threshold applied to every hostel may generate unnecessary alarms because buildings differ in size, facilities, and resident population.

Energy-saving actions should be evaluated using comparable periods before and after implementation. The comparison should consider occupancy, weather, academic schedules, and any changes in equipment or meter coverage. Where possible, the study should report both total energy and an occupancy-adjusted measure. This makes it easier to distinguish a genuine improvement in energy efficiency from a reduction caused simply by fewer residents being present.

XI. DATA QUALITY AND REPRODUCIBILITY

For the analysis to be repeatable, the dataset version, source, download date, selected files, units, time zone, and preprocessing decisions should be recorded. The cleaning procedure should state how duplicate timestamps, missing intervals, implausible values, and meter resets were handled. Charts should include units, time ranges, and sample counts so that readers can understand what is being compared. If contextual variables are merged from institutional records, the matching key and any unmatched records should also be reported.

The final report should separate three types of statements: measurements directly observed in the dataset, interpretations supported by those measurements, and possible explanations that require additional evidence. This distinction prevents the EDA from overstating what the available data can establish. It also provides a clear path for future data collection, such as adding occupancy logs, academic-calendar labels, outdoor weather records, and sub-meter readings.

A. Reproducible reporting

Every figure should state the variable, unit, observation period, and aggregation level. The number of valid observations should be reported alongside summary statistics, particularly when missing values cause different analyses to use different subsets of records. A short data dictionary should define each field and identify whether it is measured directly, derived during preprocessing, or merged from another source. These practices allow another analyst to reproduce the workflow and understand the limits of each result.

B. Recommended validation

Before drawing management conclusions, the main patterns should be checked against the original meter records and, where available, a second source such as facility logs. A sample of timestamps can be traced from raw readings through cleaning and aggregation to the final chart. Any change in meter coverage, building operation, or recording frequency should be documented because it can create an apparent trend that is unrelated to actual energy behaviour.

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